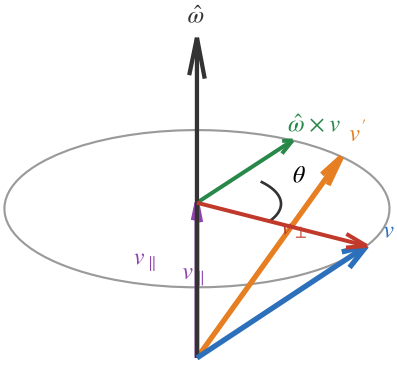
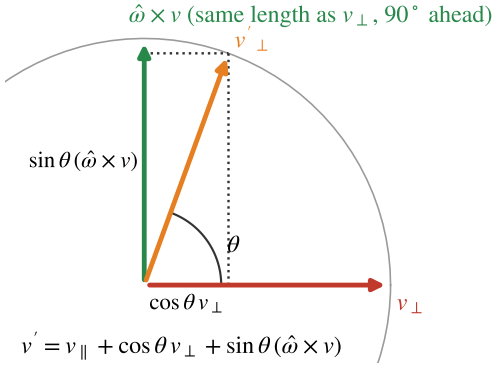
(a) only the perpendicular part rotates;
it sweeps a circle around the axis



(b) view down $\hat{\omega}$: a 2-D rotation inside the plane $\perp \hat{\omega}$



$$v' = v_\parallel + \cos\theta\, v_\perp + \sin\theta\,(\hat{\omega} \times v)$$